

AI Based Smart System Development (Real Time Hardware Implementation)

Week – 1 Project Assignment-1

- 1) Design and implement a **Smart Parking Barrier and Status Indicator System** using an ESP32, a potentiometer, an RGB LED, and a servo motor.

The potentiometer will simulate the number of available parking spaces.

Based on the potentiometer position:

- ✓ The RGB LED indicates parking availability.
- ✓ The servo motor acts as a parking gate barrier.
- ✓ The system displays the parking status and gate position on the Serial Monitor.

Potentiometer Range	Parking Status	RGB LED	Servo Position
0 – 1365	Parking Full	Red	0° (Gate Closed)
1366 – 2730	Limited Space	Yellow	90° (Half Open)
2731 – 4095	Parking Available	Green	180° (Gate Open)

- 2) Water management is one of the most important challenges in modern smart homes and industries. Water tanks must be monitored continuously to avoid overflow and dry-running conditions.

Design and implement a **Smart Water Tank Controller** using an ESP32 Development Board, a Potentiometer, an RGB LED, and a Servo Motor.

In this project:

- ✓ The Potentiometer simulates the water level sensor.
- ✓ The Servo Motor simulates the water inlet valve.
- ✓ The RGB LED indicates the current water level status.
- ✓ The ESP32 continuously monitors the water level and automatically controls the valve position.

The system should classify the water level into three zones:

Water Level	Tank Status	RGB LED	Servo Valve
0–33%	LOW LEVEL	Red	Fully Open (180°)
34–66%	MEDIUM LEVEL	Yellow	Half Open (90°)
67–100%	FULL TANK	Green	Closed (0°)

Due Date of Submission: 29/06/2026